

SIDInspector: A Mapping-First Diagnostic Resource for Semantic-ID Tokenizers

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Abstract

Semantic-ID (SID) tokenizers are increasingly released as item-to-code mappings that define the address space for generative recommenders. Yet public mappings rarely share an inspection interface: before training a generator, users must separately check catalog coverage, full-code aliasing, prefix-co-occurrence alignment, tail capacity, and prefix fan-out. We present SIDInspector, a mapping-first diagnostic resource for SID tokenizer artifacts. SIDInspector defines a small adapter contract over item mappings, metadata, interactions, and optional traces; validates the contract; and reports mapping-level probes for utilization, aliasing, neighborhood alignment, popularity allocation, and structural cost, with hooks for temporal churn and trace-based checks.

SIDInspector produces inspectable profiles, not a leaderboard. On a 23,742-item Musical worked example, a controlled GRID/RQ-KMeans-style feature-text export has 3,749 unique full codes and a 0.977 full-code aliasing rate, while a bounded ReSID/GAOQ export is aliasing-free in its exported mapping. The strongest prefix-co-occurrence alignment, however, comes from a deterministic category-prefix control rather than either learned export row (0.447 versus 0.154 and 0.055–0.080). This separates addressability from behaviorally meaningful prefixes. Auxiliary checks replicate the diagnostic direction on All_Beauty, add a real Sports GRID export with complete joins, and show that D3 is positively associated with candidate recall under a fixed train-only reranker. Mechanism probes further separate raw aliasing volume, interaction-qualified aliasing risk, head-to-tail capacity, and realized prefix cost. SIDInspector is not a new tokenizer or a benchmark suite; it makes SID tokenizer exports auditable before downstream generator training.

CCS Concepts

• **Information systems** → **Recommender systems**; *Retrieval models and ranking*.

Keywords

semantic IDs, generative recommendation, tokenizer artifacts, diagnostic evaluation, recommender systems

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1 Introduction

Semantic-ID (SID) tokenization has become a practical interface for generative recommendation. Systems such as TIGER map items into discrete code sequences so that a sequence model can generate candidate identifiers rather than score every catalog item directly [21]. Once exported, that item-to-code mapping becomes the address space exposed to the generator. Recent work has expanded this space quickly: residual quantization frameworks, recommender-native encoding, collaborative tokenization, differentiable tokenization, non-uniform capacity, qualified collisions, variable-length interfaces, and drift-aware refreshes all change the artifact that downstream users must inspect before generator training [3, 6–8, 11, 15, 25, 31]. This paper treats those exports as the object of inspection: the goal is not to reimplement the literature, but to put item-to-code mappings under one contract.

No comparable inspection layer exists for these exports. A tokenizer repository may expose checkpoints, item mappings, codebooks, or only intermediate feature files. Downstream users still need to know whether all catalog items have valid codes, whether many items share a full code, whether prefixes reflect user co-occurrence rather than only metadata taxonomy, whether tail items receive address capacity, and whether the prefix structure creates large fan-out. These checks are usually reimplemented ad hoc inside each paper, making evidence hard to compare across codebook usage, downstream ranking, and qualitative examples.

Aggregate ranking metrics remain necessary, but they do not expose the address space itself. Prior recommender tooling has argued for behavioral tests beyond NDCG-style aggregates [4]; SID tokenizers add a concrete resource problem: the item mapping should be validated and profiled before a full generator consumes GPU time. Underused codebooks, repeated full codes, behaviorally weak prefixes, tail compression, and large prefix fan-out can be hidden by a downstream score, yet they are direct properties of the tokenizer export.

We present SIDInspector, a diagnostic/interface resource for inspecting item-to-SID tokenizer artifacts before or alongside downstream recommendation evaluation. SIDInspector is deliberately neither a new tokenizer nor a RecBole-style coverage benchmark. It asks a narrower and reusable question: given any tokenizer artifact that can emit item-level code sequences, what can be said about its capacity use, aliasing, behavioral alignment, head-tail allocation, and structural cost before retraining a full generator?

The contribution is an interface, a probe suite, and calibrated evidence. The adapter contract covers SID assignments, item metadata, interaction histories, and optional generator outputs; validators reject incomplete or ambiguous rows before metrics are reported. D1–D5 report utilization, aliasing, neighborhood alignment, popularity

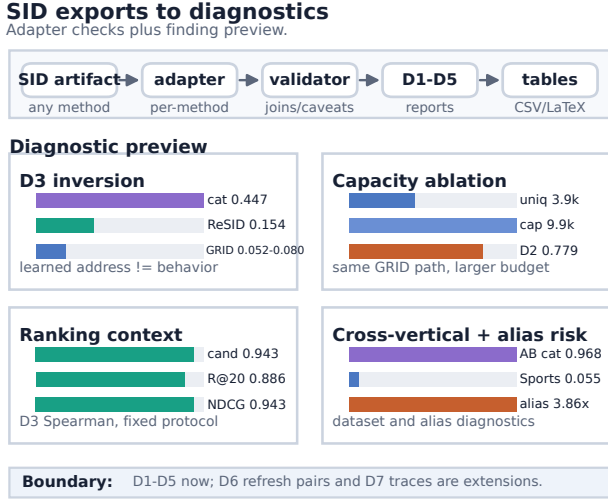


Figure 1: SIDInspector workflow and finding preview. Top: adapter and validator normalize a SID artifact before D1–D5 reports. Bottom: preview panels show the D3 inversion, capacity ablation, ranking context, and portability/aliasing checks detailed in Tables 2 and 3.

allocation, and structural cost, with D6 churn support for continual-tokenizer settings. Worked examples and controlled mechanism probes show that full-code aliasing pressure, interaction-qualified aliasing risk, collaborative-prefix recovery, tail capacity, and structural prefix fan-out are separable. In the bounded Musical example, a category-prefix control recovers more co-occurrence prefix neighbors than the learned export rows, so learned addressability and behavioral prefix alignment require separate inspection. Auxiliary vertical, fixed-reranker, and portability checks support this finding without turning the paper into a method leaderboard [5].

2 Artifact Interface and Validation

SIDInspector treats a tokenizer export as an auditable input, not as an opaque component of a trained recommender. The minimum input is a table `sid_assignments(item_id, sid_0, ..., sid_L)`. Optional tables provide metadata, interactions, refresh pairs, and generator outputs. This mapping-first contract lets adapters normalize heterogeneous repositories while keeping the audit independent of a particular training loop. Figure 1 summarizes the workflow and previews the diagnostic signals SIDInspector reports.

Artifact contract. The required contract has three checks: each row has a stable item key, every SID level is a discrete token, and the diagnostic item universe is joinable to metadata and interaction slices. Metadata enables semantic/category slices; interactions enable collaborative-neighborhood and head-tail diagnostics. The generator_outputs table is optional because many public token releases expose item codes but no trained generator traces.

Adapter specification. An adapter is a thin export layer, not a retraining recipe. It must emit `sid_assignments` with stable item

keys, a method/dataset label, and one column per SID level. If available, it also emits metadata, interactions, refresh-pair snapshots, or generator traces using the same item keys. The validator checks key uniqueness, missing mappings, depth consistency, join coverage, and declared evidence role before any row can enter the audit tables. The minimal schema is deliberately small:

$\text{item_id, method, dataset, sid_0, ..., sid_L} \rightarrow \{D1, ..., D5\}$.

Interactions add D3/D4 slices; paired mappings add D6; generator traces add D7. This makes a new tokenizer easy to attach without rewriting D1–D5. Current D1–D5 reports use a rectangular code table. Variable-length tokenizers can export padded or masked levels plus realized length; their serving behavior remains structural D5 evidence unless generator traces are available.

Terminology. A *diagnostic probe* is a named metric family: D1 utilization, D2 aliasing, D3 neighborhood alignment, D4 popularity allocation, D5 structural cost, D6 temporal churn, or D7 generation traces. A *controlled mechanism probe* is an input row designed to activate one known mechanism and check whether the diagnostic probe reacts. Current paper evidence uses D1–D5. D6 is implemented as an extension for paired refresh mappings and has one repository smoke result; D7 is only an interface hook until generator outputs or beam traces are supplied.

Claim scope. SIDInspector separates rows by what they can support. A named-method row comes from a public method path and can support named-method artifact claims. A controlled mechanism probe is deliberately synthetic or method-inspired and tests whether a diagnostic reacts to a known mechanism, but it is not named-method coverage. Optional rows exercise extensions such as drift or non-Amazon schemas. Literature-only rows motivate the design space but are not loaded as runnable evidence.

Validation before metrics. Adapters cannot enter metric tables until they pass item-count agreement, missing-code, duplicate-key, depth-consistency, and provenance checks. This guard is necessary because public SID repositories may expose checkpoints, item mappings, codebooks, intermediate feature files, or only partial releases. The manifest records each row’s evidence role and caveats before any diagnostic number is emitted.

Prior recommender resources such as RecList and Elliot motivate diagnostic and reproducible evaluation beyond a single aggregate metric [1, 4]. SIDInspector applies that resource logic to a newer artifact boundary: the item address space consumed by generative recommenders.

3 Diagnostic Probes and Evidence Roles

SID papers now optimize different artifact properties: learned item indexing and TIGER-style generative retrieval established the interface [9, 21], while later work stresses collaborative alignment, end-to-end or differentiable tokenization, representation-source sensitivity, code-assignment diversity, semantic-collaborative disentanglement, collision qualification, variable-length or asymmetric interfaces, drift, and deployment surfaces [2, 3, 7, 8, 10, 12, 16–18, 20, 24, 28, 30, 31]. The literature space is broader than the artifacts that SIDInspector currently executes. Table 1 summarizes what the released artifact ships rather than counting baselines. The

Table 1: Evidence catalog for the released artifact.

Row	Type	Items	Seeds	D coverage
GRID ft	named (A)	23,742	3	D1–D5
GRID ft-cap	named (A, capacity ablation)	23,742	1	D1–D5
RQ-min ref	reference adapter	23,742	1	D1–D5
ReSID [†]	named (B)	23,742	1	D1–D5
Cat-prefix, Pop-bal, Hash-coll	controls	23,742	n/a	D1–D5
Mechanism probes (3)	controlled	synth.	n/a	D2,D4,D5
All_Beauty, Sports, MovieLens, portability/extension DACT		varies	varies	D1–D5, D6

taxonomy guides future adapters; it is not presented as reproduced-method coverage.

D1–D5 mapping-level probes. D1, *utilization*, reports per-level usage, prefix counts, imbalance summaries, and dead or underused code indicators. D2, *aliasing*, measures duplicate full SIDs and duplicate prefixes; it is an aliasing profile, not a causal harm estimate. A bounded interaction-qualified mechanism probe is included because collision-aware work argues that duplicate-code assignments are not uniformly harmful [8]. D3, *neighborhood alignment*, asks whether prefix neighborhoods recover item co-occurrence neighbors, keeping metadata purity as auxiliary context rather than as collaborative quality. D4, *popularity allocation*, splits items by popularity and measures whether full-code capacity and prefix structure are allocated differently across head, mid, and tail items. D5, *structural cost*, reports SID length, prefix fan-out, duplicate codes, and trie-like expansion pressure. D5 is not serving latency; long, variable-length, or asymmetric SID methods make that distinction important [3, 10, 23, 26, 29].

Extension probes. D6, *temporal churn*, computes mapping changes between tokenizer refreshes for drift-aware settings [2, 6]. It is implemented as optional refresh-pair evidence, not a required main-paper claim. D7, *generation traces*, covers invalid paths, next-token entropy, and duplicate generated candidates. D7 requires generator-output tables or beam traces and is left as an interface hook in the current release.

4 Worked Examples and Probe Evidence

The evaluation asks whether one adapter contract can turn heterogeneous SID exports into comparable profiles. GRID supplies a released GRID/RQ-KMeans-style module; ReSID supplies processed Musical artifacts and a bounded GAOQ export path. Sanity rows and controlled mechanism probes calibrate which differences are structural, behavioral, or capacity-driven.

Worked-example protocol. The case study uses the same 23,742 Musical items: a controlled GRID/RQ-KMeans feature-text export from ReSID text features, plus a bounded ReSID FAMA-to-GAOQ export. This supports same-item artifact comparison, not faithful reproduction of TIGER, GRID, or full ReSID training.

Table 2 puts the RQ-style rows together so capacity and aliasing changes can be read before the ReSID contrast and the sanity controls. The GRID ft row is labeled as feature-text because it uses processed feature text from the ReSID artifact path and the released residual k-means module, not a faithful raw-text TIGER reproduction. In this controlled setup, the row exposes high aliasing pressure

Table 2: SID diagnostics on Amazon Reviews 2023 Musical Instruments. Rows follow the diagnostic path: RQ-style variants, ReSID contrast, then controls. Bold rows are named-tokenizer export paths; italics are controls.

Artifact	Unique SIDs	D2	D3	D4	D5
GRID ft	3,749	0.977	0.055	0.370	64/3.4k/3.7k
GRID ft-cap	9,874	0.779	0.080	0.639	32/9300/9874
RQ-min ref	17,247	0.440	0.065	0.883	32/2368/17.2k
ReSID[†]	23,742	0.000	0.154	1.000	32/1280/23.7k
<i>Cat-prefix[†]</i>	23,742	0.000	0.447	1.000	30/83/313/23.7k
<i>Pop-balanced</i>	22,707	0.086	0.303	0.962	4/1024/22.7k
<i>Hash-collide</i>	256	1.000	0.004	0.032	256/256/256/256

Table 3: Controlled mechanism probes. Each compact contrast activates a known artifact pressure and checks whether the targeted diagnostic separates it.

Probe	D	Without probe	With probe
Qualified aliasing	D2	hash 1.19x	co-occur 3.86x
Capacity budget	D1/D4	head 1.000	tail 0.028
Variable depth	D5	max-depth 12,010	active 7,914

and limited tail capacity. The bounded ReSID/GAOQ row exports one full code per item, yielding zero full-code collisions and higher D3–L1 collaborative prefix recovery. The daggered rows mark a structural floor: collision-free full codes are not by themselves evidence of better recommendation quality. The capacity-matched GRID row reopens the strongest alternative explanation: when the same feature-text path uses a larger 32/1280/1280 budget, unique full codes rise to 9,874 and D2 aliasing drops to 0.779, but the artifact still does not approach the structurally item-unique ReSID or category-prefix rows. Thus capacity explains part of the original GRID pressure. The ablation is not a full item-unique leaf match, and additional capacity could reduce aliasing further; the contrast is therefore an artifact profile rather than a method ranking. The RQ-min row serves a different role: it is a minimal residual-quantization reference adapter that passes the same validator on the full Musical item universe. It shows that SIDInspector accepts an independent SID-generation code path, but it is not third named-method coverage and does not reproduce TIGER, GRID, or ReSID.

Diagnostic findings. Table 2 separates addressability from behavioral prefix alignment. The ReSID row is structurally item-unique; the category-prefix control is also alias-free because it uses metadata categories as early levels and a within-prefix item index as the leaf. It is a taxonomy control rather than a learned tokenizer. D3 exposes the surprising part: the non-learned category-prefix row recovers more level-1 co-occurrence neighbors (0.447) than ReSID (0.154) or either GRID row (0.055–0.080). The point is not that category IDs are better tokenizers, but that learned or item-unique addresses do not guarantee behaviorally aligned prefixes. Direct mapping inspection makes that failure mode visible.

The capacity ablation gives an intervention-style check: one failure mode can improve while another persists. Increasing the GRID feature-text budget to 32/1280/1280 raises unique full codes from the 3.7–4.0k range to 9,874 and reduces D2 aliasing from 0.977 to 0.779. This weakens a simple “too little capacity” objection, but it does not make the row item-unique. The RQ-style rows

expose a sharper diagnostic dissociation: D2 aliasing falls from 0.977 (GRID ft) to 0.779 (GRID ft-cap) and 0.440 (RQ-min), while D3 stays in the narrow 0.055–0.080 range. Capacity and aliasing can improve without producing behaviorally aligned prefixes, which is why these probes are reported separately. The controlled probes in Table 3 give the same separation under simpler inputs: qualified aliasing separates co-occurrence aliases from hash aliases, capacity probes separate head and tail allocation, and variable-depth probes separate maximum-depth from realized prefix cost.

The D3 pattern is not confined to Musical. On an All_Beauty 20k vertical, a coarse category-prefix control reaches D3-L1 0.968 while GRID/RQ-KMeans feature-text rows are 0.081, 0.087, and 0.090 across seeds 42–44. The coarse-category flag is essential: this is diagnostic portability evidence, not a claim that taxonomy IDs are better tokenizers. The unusually high All_Beauty value is itself part of the diagnostic signal: D3 distinguishes a vertical where the available coarse taxonomy is strongly aligned with observed co-occurrence from Musical, where the same category-prefix control is only moderately aligned. A plausible explanation is that Beauty co-purchase sessions often stay within narrow product-type and routine categories, while Musical accessory and instrument co-purchases can cross coarser taxonomy boundaries. For external grounding, a 5,000-user Musical check uses SID prefixes only as candidate sets and applies the same train-only co-occurrence/popularity reranker to all six rows; D3 correlates with candidate recall (Spearman 0.943) and fixed-reranker Recall@20 (0.886). A bounded All_Beauty temporal-LOO replication is consistent in direction across four rows: D3 is monotonic with candidate recall and fixed-reranker Recall@20, but the small row count and constructed split keep it as supplementary directional evidence. The GRID export path also extends to Sports: a real 20k GRID export has complete metadata and interaction joins, D3-L1 0.055, duplicate-SID rate 0.592, and D5 prefixes 128/7986/8165. Sports proxy rows remain supplement-only and do not add named method coverage. DACT D6 and MovieLens schema rows exercise extension and portability paths, but the main paper claim remains the Musical worked example.

5 Artifact Availability and Limits

The repository contains the SIDInspector Python package, adapter scripts, metric runners, generated CSV/Markdown/L^AT_EX tables, provenance logs, BibTeX/source-audit notes, an MIT license, and a reviewer quickstart. The anonymous review artifact is available at <https://anonymous.4open.science/r/sidinspector-9BB2>, with a clean-checkout verifier for the published tables and figure. Large local artifacts are kept out of git and described through manifests so reviewers can separate public reproducibility assets from local caches.

Users can apply SIDInspector in three modes: downstream researchers flag incomplete, aliased, or behaviorally weak artifacts before generator training; method authors start from the minimal adapter template and receive D1–D5 reports; and reviewers run the clean-check verifier against published CSVs. Runtime notes separate lightweight D1/D2/D4/D5 preflight checks from the interaction-dependent D3 step. These signals do not replace Recall@K or NDCG; they make tokenizer artifacts inspectable before a full benchmark or generator-training run.

The evidence supports a resource-interface claim, not broad method coverage. The GRID Musical row is a controlled feature-text export rather than an end-to-end TIGER or GRID reproduction; ReSID is the bounded Musical export; RQ-min is a local reference adapter; the Sports balanced GAOQ path was stopped after constrained k-means became CPU-bound; and CARD remains a fallback/probe path rather than faithful CARD reproduction [25]. D2 is not causal collision harm; D3 uses prefix-neighborhood and reranker checks, not trained-generator ranking; D5 is structural cost rather than measured latency; D6 is optional churn evidence; and D7 requires generator traces absent from the current rows.

New adapters emit the same normalized tables, declare their evidence role, and record feature or checkpoint provenance. A named tokenizer improves coverage only after item-level codes pass the validator and join to metadata or interactions. A controlled mechanism probe improves diagnostic calibration but does not count as named-method coverage; a generator trace activates D7. Future versions should add causal collision-harm validation, broader faithful adapter coverage, generator-output diagnostics, and unified search/recommendation checks [14, 19]. Industrial SID deployment and ranking studies further motivate treating these artifact properties as engineering objects [12, 13, 22, 27].

GenAI Usage Disclosure

Generative AI tools assisted with language editing and code debugging. All experimental results were produced by author-written scripts and verified against saved artifacts.

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